lookatit

ORCA Trajectory & Geometry Visualizer

User Manual & Technical Documentation

September 7, 2026

1 Overview

lookatit is an interactive Python web dashboard engineered for rapid post-processing of ORCA relaxed surface scan trajectories and single-structure coordinate files. By parsing XYZ coordinate blocks directly, **lookatit** synchronizes an interactive WebGL 3D molecular viewport with dual-axis relative energy and geometrical parameter plots.

2 Core Capabilities

- **Robust Coordinate Parsing:** Supports multi-frame ORCA `.trj/.xyz` files as well as static single-frame `.xyz` files.
- **Automatic Energy Normalization:** Extracts Hartree electronic energies from comment headers and converts them to relative $kcal/mol$.
- **Smart Geometric Tracking:** Computes structural metrics dynamically based on the number of provided 0-indexed atom indices:
  - **2 Indices (e.g., `0, 1`):** Bond Distance ($\AA$).
  - **3 Indices (e.g., `0, 1, 2`):** Bond Angle (°) centered at atom 2.
  - **4 Indices (e.g., `0, 1, 2, 3`):** Dihedral Angle (°).
- **Synchronized Inspection:** Scrubbing through scan steps simultaneously updates the 3D atomic render, highlighted measurement cylinders, dynamic text tags, and plot cursors.

3 Installation & Quickstart

3.1 1. Virtual Environment

```
mkdir lookatit && cd lookatit
python3 -m venv venv
source venv/bin/activate
```

3.2 2. Dependencies

```
pip install streamlit py3Dmol matplotlib numpy
```

3.3 3. Launch Application

```
streamlit run lookatit.py
```

4 Workflow Guide

1. **Upload File:** Load an ORCA `.trj` or `.xyz` file in the sidebar.
2. **Define Measurement:** Input comma-separated 0-indexed atom numbers (e.g., `0, 1` or `5, 12, 15, 18`).
3. **Analyze:** Step through frames using the main slider to inspect relative energy profiles and structural parameter shifts.